2 Project Proposal

Project Idea

The Smart Liquid Tank Monitoring and Measurement System aims to design and build a system that monitors the liquid level, temperature, and weight inside a tank using sensors connected to an Arduino Uno.

The system is useful in industrial applications, water storage tanks, factories, and liquid processing systems. It helps monitor the amount of liquid inside the tank, detect high or low liquid levels, and measure temperature changes. It can also help prevent tank overflow by providing a warning when the liquid reaches a predefined level.

2.2 Main Specifications

The system will use the following sensors and components to collect and display measurements:

1 Ultrasonic Sensor (HC-SR04): Measures the distance between the sensor and the liquid surface to calculate the liquid level and remaining empty space inside the tank.

2 Temperature Sensor (DS18B20): Measures the liquid temperature in degrees Celsius (°C) to monitor temperature changes and support safe operation.

3 Load Cell with HX711 Module: Measures the weight of the tank's contents to estimate the amount of liquid stored inside it.

4 Arduino Uno**:** Receives sensor readings, processes the data, and controls the system.

5 Buzzer and LED**:** Provide a warning when the liquid level exceeds a predefine

Item List

1 Arduino Uno.

2 HC-SR04 ultrasonic sensors.

3 DS18B20 waterproof temperature sensor.

4 Load cell.

5 HX711 load cell amplifier module.

7 Small liquid tank or plastic container.

8 Buzzer.

9 LED and current-limiting resistor.

10 Breadboard.

11 Jumper wires.

12 USB cable or suitable power supply.

13 Calibration weights.

14 Measuring container for testing and calibration.

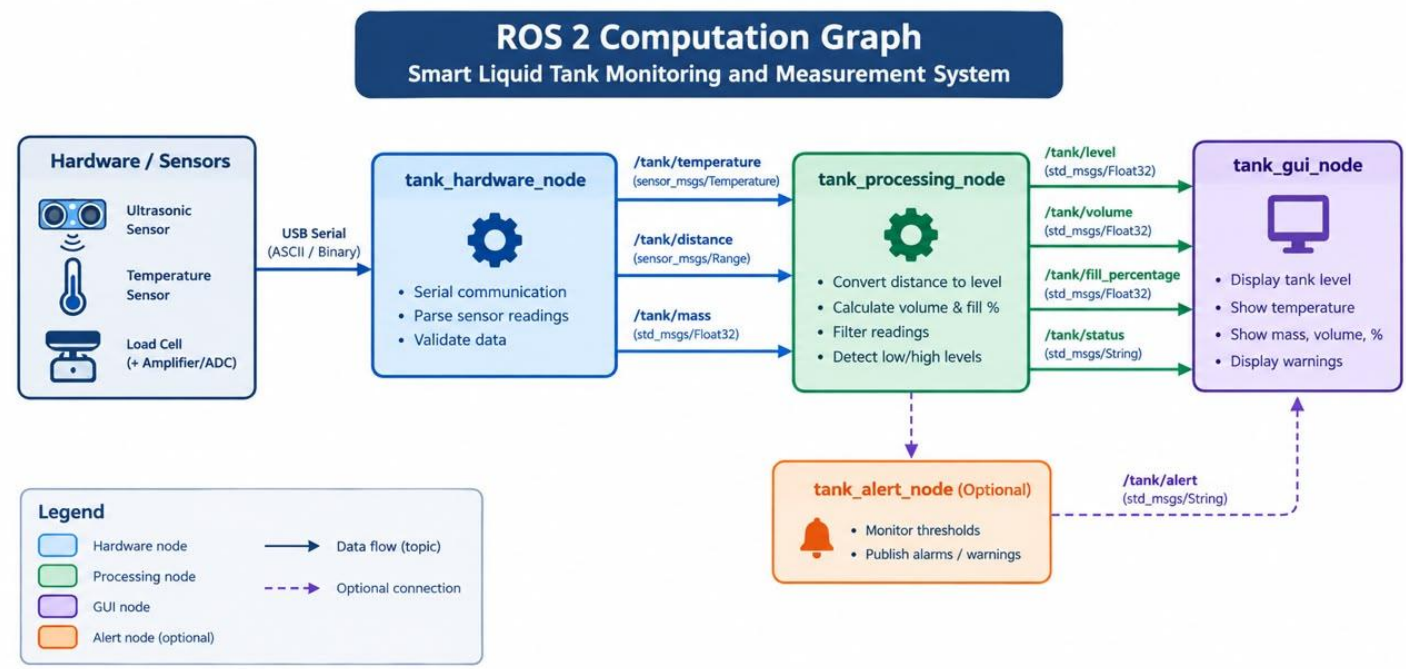
Methodology

the project will be implemented by connecting the ultrasonic sensor, temperature sensor, and load cell to the Arduino Uno. The ultrasonic sensor will measure the distance from the sensor to the liquid surface, allowing the liquid height to be calculated using the known tank dimensions. The DS18B20 sensor will measure the liquid temperature, while the load cell and HX711 module will measure the weight of the liquid inside the tank.

The Arduino will process the readings and display the results on the Laptop screen. A buzzer and LED will be used to indicate when the liquid level reaches a predefined warning limit.

Finally, the system will be tested using different liquid levels and known weights. The measured values will be compared with reference measurements to evaluate the accuracy and reliability of the system.

Workflow



3. ROS Topics and Message Types

3.1. Sensor Inputs

tank_hardware_node → tank_processing_node

Topic	Message Type	Purpose
`/tank/temperature`	`sensor_msgs/msg/Temperature`	Liquid temperature (°C)
`/tank/distance`	`sensor_msgs/msg/Range`	Ultrasonic distance (m)
`/tank/mass`	`std_msgs/msg/Float32`	Measured mass (kg)
`TBD`	`Custom message`	Raw sensor data from microcontroller

3.2. Processed Outputs

tank_processing_node → tank_gui_node

Topic	Message Type	Purpose
/tank/level	std_msgs/msg/Float32	Liquid height (m)
/tank/volume	std_msgs/msg/Float32	Liquid volume (L)
/tank/fill_percentage	std_msgs/msg/Float32	Fill level (%)
/tank/status	std_msgs/msg/String	Status (e.g., NORMAL, LOW_LEVEL)

3. Alerts

tank_alert_node → tank_gui_node

Topic	Message Type	Purpose
/tank/alert	std_msgs/msg/String	Alarm / warning (optional)